

Clear-sky convergence and the origin of tropical congestus clouds

FRANCISCO E. SPAULDING-ASTUDILLO ¹ AND JONATHAN L. MITCHELL^{2,1}

¹*Department of Earth, Planetary, and Space Sciences, University of California, Los Angeles
595 Charles E Young Dr E, Los Angeles, CA 90095, USA*

²*Department of Atmospheric and Oceanic Sciences, University of California, Los Angeles
520 Portola Plaza, Los Angeles, CA 90095, USA*

ABSTRACT

Congestus clouds, characterized by their vertical extent into the middle troposphere, are widespread in tropical regions and play an important role in Earth’s climate system by their contribution to cloud radiative forcing, atmospheric humidification, and surface rainfall. However, their spatial distribution – in particular the abundance of stratiform clouds sourced by the outflow from congestus cloud tops – is inaccurately captured by state-of-the-art climate models, suggesting that fundamental questions regarding their formation, dynamics, and climate impact remain unanswered. Here, we demonstrate how congestus cloud tops form by detraining preferentially at altitudes between 5–6 km and why they are more prevalent in dry mid-tropospheric conditions. Convective detraining maximizes at a height of 5–6 km due to a swift decline in radiative cooling in clear-sky regions. This decline is, in turn, a consequence of the non-uniform density of strong absorption lines in the water vapor rotation band. The increased prevalence of congestus clouds in drier mid-tropospheric conditions is due to stronger vertical gradients in the clear-sky cooling rate, which drives stronger outflow at 5–6 km. Our finding of a clear-sky radiative mechanism for congestus cloud top formation could be used to improve parameterizations of mid-level cloud fraction in general circulation models, which systematically underestimate this major cloud population.

Keywords: Radiative transfer (1335) — Atmospheric clouds (2180) — Atmospheric dynamics (2300)

1. INTRODUCTION

There is a clear need to better understand cloudiness in varied environmental and dynamical conditions on Earth because clouds are crucial to the atmospheric energy and moisture budgets. In fact, clouds remain the largest source of uncertainty in estimates of Earth’s climate sensitivity to anthropogenic carbon dioxide emissions (Ceppi & Nowack 2021). The range of spatial and temporal scales over which clouds form and evolve in response to the complex interaction of cloud microphysics with radiation, convection, and large-scale dynamics make clouds especially challenging to represent faithfully in climate models. Models that do not resolve cloud-scale motions (e.g., general circulation models) rely on parameterizations or “schemes” to calculate cloud fraction. Parameterizations compute the unresolved, sub-grid scale variability of a climate variable in terms of resolved, grid scale quantities (Smith 1990). Even so-called cloud-resolving models (CRMs) employ parameterizations for the cloud microphysics, which occur at spatial scales several orders of magnitude smaller than the finest model resolution. In the simplest cloud schemes, a unique relationship between the grid scale relative humidity and sub-grid scale cloudiness is assumed, thus enabling the latter to be diagnosed from the former. This longstanding assumption is backed by considering the statistics of unresolved sub-grid scale fluctuations in relative humidity. Spatial variations in moisture and temperature over sub-grid scales yield regions where the humidity is greater than or less than the local saturation value (Tompkins 2005), which should result in fragmented patches of cloud and clear-sky that are unresolved by the climate model. As the mean relative humidity increases, it follows that a greater fraction of the grid cell should be cloudy. This argument is entirely valid

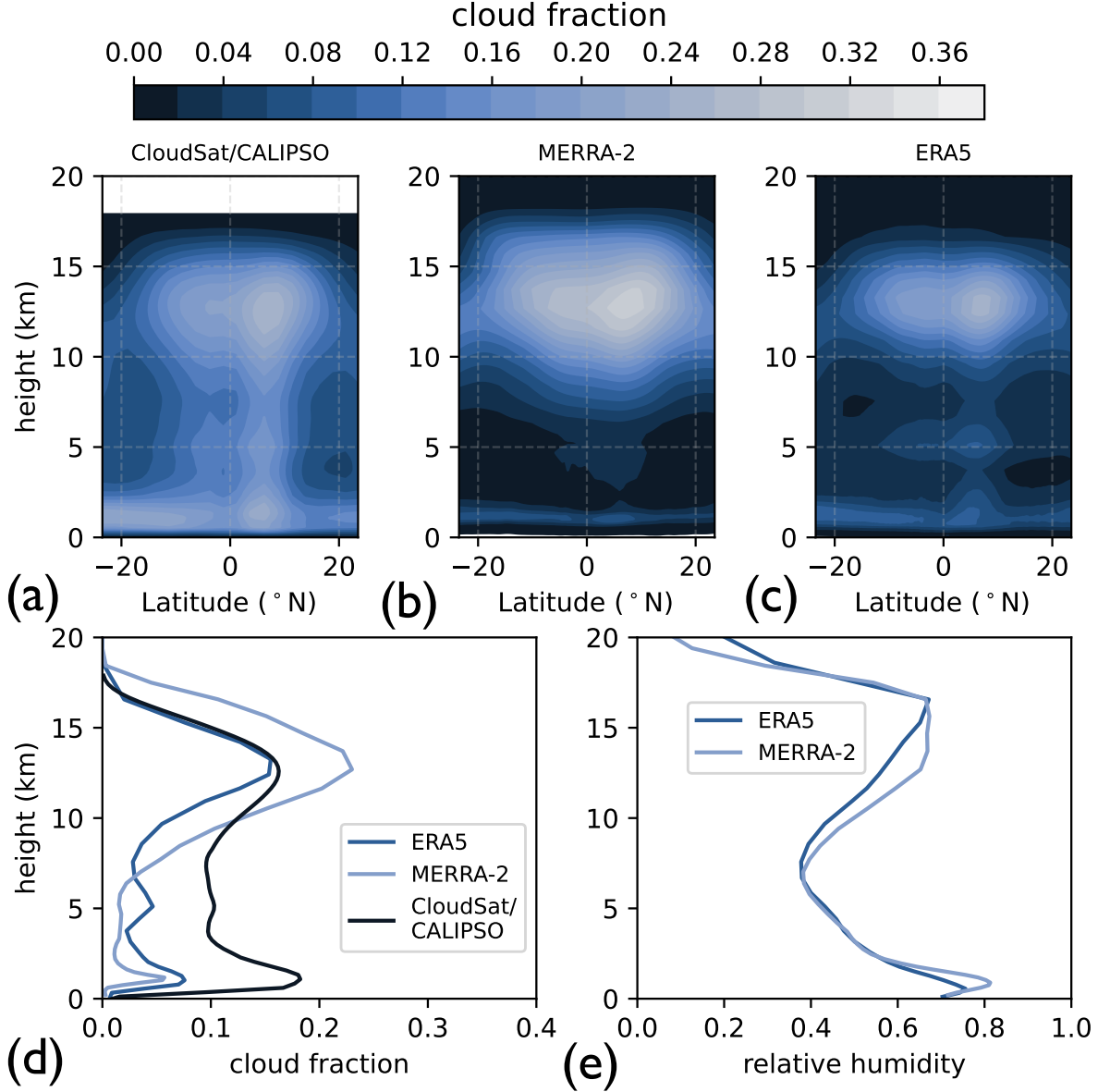


Figure 1. Zonal-mean average of cloud fraction in the tropics over 2006-2019 from (a) combined CloudSat/CALIPSO observations (Bertrand et al. 2023), (b) the MERRA-2 re-analysis (Global Modeling and Assimilation Office 2015), and (c) the ERA5 reanalysis (Hersbach et al. 2023). Tropical-mean (d) cloud fraction and (e) relative humidity from various products.

for stratiform clouds, but it has been long recognized that the same argument likely does not extend to convective clouds (Tompkins 2005), which form and evolve in response to different physical processes (Sundqvist 1988). The widespread adoption of relative humidity-based schemes in climate models has made it commonplace to interpret high cloudiness within the model as a symptom of high relative humidity (Spaulding-Astudillo & Mitchell 2023).

Tropical clouds, i.e. those found within approximately 23.5 degrees of the equator, appear in three distinct cloud populations or “modes”: shallow cumulus, mid-level congestus and deep cumulonimbus (Johnson et al. 1999). A simple way to demonstrate the existence of the three modes is by inspection of the tropical-mean cloud fraction from CloudSat/CALIPSO observations (Figure 1a; Bertrand et al. 2023), which reveal three peaks in the vertical distribution of cloudiness. Shallow cumulus, also known as fair-weather clouds, form at the top of the boundary layer and cool the climate by reflecting sunlight (Cesana et al. 2019; Albrecht et al. 2019). Precipitating cumulonimbus rise out of the boundary layer and detrain near the tropopause, forming anvils as the cloudy air spreads out laterally (Hartmann et al.

2018; Romps 2014). The detraining anvils are also associated with the production of thin, high cirrus, which have a greenhouse warming effect on the climate (Sassen et al. 2009). Congestus clouds extend well above shallow cumulus but are less vertically developed than deep cumulonimbus, terminating in the mid-troposphere (Wall et al. 2013; Mechem & Oberthaler 2013). Thin, mid-level clouds with a cooling effect are ubiquitous in the tropics and are associated with convective detraining at congestus peaks (Bourgeois et al. 2016). Unlike the other modes, congestus cloud tops form where the relative humidity tends to be low (Figure 1e). They play an important role in the humidification of the middle troposphere (Hohenegger & Stevens 2013) and are responsible for 25-40% of the total rainfall in the tropics (Johnson et al. 1999; Petty 1999). At the same time, global climate model simulations and reanalyses¹ are notoriously poor at representing the congestus mode. This can be seen by comparing CloudSat/CALIPSO observations (Figure 1a; Bertrand et al. 2023) to the MERRA-2 (Figure 1b; Gelaro et al. 2017) and ERA5 (Figure 1c; Hersbach et al. 2020) reanalysis products (Figure 1d compares the horizontal averages of all three). Reanalyses are produced using global climate model simulations that are nudged towards real observations, so they in part rely on parameterizations for cloud physics that the model does not resolve. Figure 1 reveals the well-known problem (Miao et al. 2019; Bodas-Salcedo et al. 2008) that reanalyses (and climate model simulations in general) often underestimate the mid-level cloud fraction and/or fail to resolve the mid-tropospheric peak observed at ~ 5 km in CloudSat/CALIPSO (Figure 1a,d). A goal of this paper is to illuminate the origin of the mid-tropospheric (congestus) peak in tropical cloud fraction, and thereby pave the way toward better representation of this important cloud mode in climate simulations and reanalysis.

Tropical congestus clouds reach their peak altitude at 5-6 km, with the spread indicative of the typical values over ocean and land areas (Wall et al. 2013). Two theories based on buoyancy arguments have been proposed to explain the height of the congestus peak: the weak stability theory and the dry-air entrainment theory. The shared premise of these theories is that some process decelerates cloudy air as it ascends, causing it to accumulate and spread out preferentially at this level (Redelsperger et al. 2002; Jensen & Del Genio 2006). The weak stability theory posits that semi-permanent and/or transient areas of weak stability in the mid-troposphere act as barriers to convective cloud development. The proposed mechanisms that promote weak stability invoke water phase changes, such as local evaporation of detrained cloud condensate (Nuijens & Emanuel 2018), sublimation cooling associated with dry air intrusions (Zuidema et al. 2006), and melting of stratiform cloud ice as it crosses the 0°C freezing level at 4-5 km in typical tropical conditions (Johnson et al. 1996). The freezing level is commonly referenced in studies of tropical congestus (Johnson et al. 1999; Jensen & Del Genio 2006; Mechem & Oberthaler 2013) because their tops are prevalent near it. The second theory, dry-air entrainment, asserts that congestus clouds peak in the mid-troposphere where the relative humidity is lowest due to buoyancy loss from mixing between the cloud and its dry environment (Brown & Zhang 1997). A strong consensus as to which mechanism is most important to congestus clouds seems to be lacking, with some arguing in favor of weak stability (Zuidema 1998; Riihimäki et al. 2012) or dry-air entrainment (Brown & Zhang 1997; Takemi et al. 2004; Jensen & Del Genio 2006; Takayabu et al. 2010; Kumar et al. 2014) or both (Johnson et al. 1999; Redelsperger et al. 2002). There is, on the other hand, concordance between observations and cloud-resolving model simulations that congestus clouds are more prevalent where the mid-level relative humidity is low (Brown & Zhang 1997; Redelsperger et al. 2002; Takemi et al. 2004; Jensen & Del Genio 2006; Kumar et al. 2014; Sokol & Hartmann 2022). These studies also find that areas with low mid-level relative humidity have fewer deep convective clouds (cumulonimbus).

A well-rounded theory of congestus cloud top formation should be able to explain (1) the 5-6 km peak in cloud fraction *and* (2) the enhancement of congestus by low mid-tropospheric relative humidity. While weak stability at, for instance, the freezing level of water may indicate why congestus clouds terminate near 5 km, it does not explain the invigoration of congestus clouds by low mid-tropospheric relative humidity. Similarly, the dry-air entrainment theory implies that congestus/cumulonimbus clouds should be more/less prevalent when the mid-level relative humidity is low, but it does not tell us why congestus clouds peak at 5-6 km, which is well below the tropospheric minimum in relative humidity at 7.5 km (Figure 1e). Of course, there may be a coordination between both buoyancy-reduction mechanisms that reproduce the observations. *Therefore, the aim of this study is to forge a new perspective of congestus cloud formation that accounts for a greater array of the observations.* Our hypothesis for the origin of congestus clouds and their invigoration builds on the so-called fixed anvil temperature (FAT) hypothesis, which we now describe.

¹ Global climate models simulate very large spatial domains and thus are distinct from cloud-resolving models, which are run over small domains at much higher resolution.

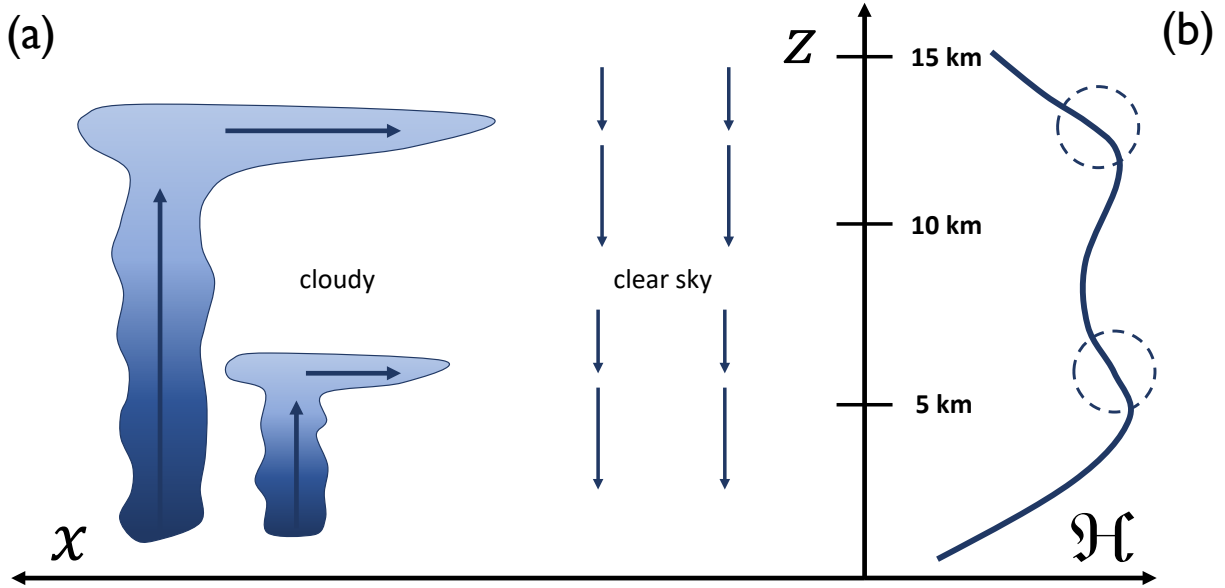


Figure 2. The radiative-convective perspective of cloud formation links the decline with altitude in clear-sky radiative cooling $\mathcal{H}$ to the transport of cloudy air into clear-sky regions, as demanded by the vertical divergence of air in clear-sky regions. Schematically (b), there are two regions where $\mathcal{H}$ decreases with height, which are associated with middle and high peaks in cloud fraction (a). The high-altitude peak at ~ 12 km is associated with cumulonimbus anvils. Our study focuses on the middle “congestus” peak at ~ 6 km.

Hartmann & Larson (2002) hypothesized a connection between the height of anvil clouds and the infrared emissivity of water vapor, which ultimately determines the ability of the atmosphere to radiatively cool. Notably, a reduction in emissivity around the 200 hPa level produces a swift decline in radiative cooling. This phenomenon is realized in cloud-resolving model simulations (Zelinka & Hartmann 2010), wherein the strongest convective outflow occurs precisely where radiative cooling declines at the fastest rate with increasing height. Given that the concentration of water vapor in the upper troposphere is primarily governed by temperature, it was hypothesized that these anvil clouds should consistently form at a specific temperature. The FAT hypothesis is generally backed by the results of observation and modeling (Kuang & Hartmann 2007; Zelinka & Hartmann 2010; Li et al. 2012). We note that a few studies have recently challenged the notion of a strictly fixed cloud-top temperature (Seeley et al. 2019a; Seidel & Yang 2022). The underlying cause of the decline in emissivity at the upper tropospheric boundary was later clarified by Jeevanjee & Fueglistaler (2020a). Employing a line-by-line radiative transfer model, they revealed that the diminishing upper-tropospheric emissivity in clear-sky regions partly stems from a decrease in the abundance of strongly-absorbing wavenumbers within the rotational band of water vapor above the 200 hPa level. Their result motivates the question: can the congestus peak be similarly understood through clear-sky infrared emissivity? On large spatio-temporal scales, the tropics are in a state of energy balance between radiative cooling and latent and sensible heating known as radiative-convective equilibrium (RCE; Jakob et al. 2019; Miyawaki et al. 2022). In RCE scenarios where the contribution from latent heating dominates, moist convection organizes into fast, narrow updrafts and slow, widespread downdrafts (Singh & O’Neill 2022). It is reasonable to assume that most of this latent heating occurs in ascending cloudy regions. Since clouds are effective at limiting the outgoing longwave radiation to space, it is also reasonable to assume that most of the radiative cooling occurs in the descending clear-sky regions. Since cloudy regions and clear-sky regions are energetically connected, the physics that governs one must also influence the other. From this simple conceptual picture, a *clear-sky* mechanism of cloud formation emerges in which cloudy air converging into clear-sky regions is connected to vertical decreases in the clear-sky cooling rate. The “clear-sky convergence” (CSC) hypothesis is at the heart of the FAT hypothesis (Seeley et al. 2019b), and its applications could be broader than previously recognized. In this study, we test the CSC hypothesis against congestus clouds.

2. A CLEAR-SKY HYPOTHESIS OF CLOUD FORMATION IN RADIATIVE-CONVECTIVE EQUILIBRIUM

We now develop a predictive model of cloud-top formation based on the CSC hypothesis. Consider an idealized tropical environment in which the temperature T is horizontally homogeneous and there is widespread subsidence in clear-sky regions and narrow ascent in cloudy regions (Figure 2a), which are linked by mass balance (Hartmann & Larson 2002; Seeley et al. 2019a). We assume that anvil clouds are generated over large horizontal areas due to the convergence of cloud water into clear-sky regions, an assumption that is supported by cloud-resolving model simulations (Zelinka & Hartmann 2010; Beydoun et al. 2021; Jeevanjee & Zhou 2022). Conservation of energy for air parcels in clear-sky regions can be expressed as

$$\rho c_p \frac{DT}{Dt} = \frac{\partial p}{\partial t} + H, \quad (1)$$

(i) (ii) (iii)

where ρ [kg/m^3] is density, c_p [$J/(kgK)$] is the specific heat of environmental air at constant pressure, p [Pa] is pressure, H [$J/(m^3s)$] is diabatic heating, and D/Dt is the total (Lagrangian, parcel-following) time derivative. Conservation requires the change in energy density of the air (i) balance subsidence warming (ii) and diabatic cooling (iii), the latter including both radiative and latent heating. If the diabatic cooling in clear-sky regions is primarily radiative and shortwave absorption is small, then H can be approximated by the infrared cooling rate $\mathcal{H}$ (by convention, positive values for cooling). Note that re-evaporation of cloud condensate in clear-sky regions is another potentially important source of local cooling that can increase the clear-sky convergence (Jeevanjee & Zhou 2022), which we neglect. In steady state (temperatures not evolving in time), (i) simplifies to a vertical advection of energy density, $\rho c_p w \partial_z T$, where w [m/s] is the vertical wind and ∂_z is the vertical derivative. We expand the pressure derivative on the right side with the chain rule, divide both sides by the heat capacity c_p , and invoke hydrostatic balance ($\partial_z p = -\rho g$ where g is the surface gravity). Using the definitions of vertical velocity $w = \partial_t z$, the lapse rate $\Gamma = -\partial_z T$, and the dry adiabatic lapse rate $\Gamma_d = g/c_p$, the clear-sky mass flux ($M_{clr} = \rho w$; positive downward) is

$$M_{clr} = \frac{\mathcal{H}}{c_p \sigma}, \quad (2)$$

where $\sigma = \Gamma_d - \Gamma$ is the static stability (positive values are stable to dry convection). If Γ and $\mathcal{H}$ were constant with height (the former implying constant σ), then M_{clr} would be constant also. However, if $\mathcal{H}$ decreases with height or σ increases with height, Equation 2 indicates a vertical divergence of mass from that layer. By mass continuity, (vertical) divergence in M_{clr} must be balanced by clear-sky (horizontal) convergence:

$$\begin{aligned} CSC &= \frac{-\partial_z M_{clr}}{\rho} \\ &= \frac{-1}{c_p \rho \sigma} \partial_z \mathcal{H} + \frac{\mathcal{H}}{c_p \rho \sigma^2} \partial_z \sigma. \end{aligned} \quad (3)$$

Positive values of CSC support a net transport of air from the cloud into clear-sky regions – i.e. detrainment – and following Seeley et al. (2019b) is specified with the units of an inverse timescale. Equation 3 shows that CSC is supported either by a decrease in radiative cooling or an increase in stability with height. In regions where the first term is positive, however, the second term is typically negative and thus reduces the net positive CSC (Seeley et al. 2019a). If we assume that the mid-tropospheric lapse rate is constant, the CSC at the congestus level would be dominated by vertical changes in the clear-sky cooling rate.

$$CSC \approx \max \left(0, \frac{-1}{c_p \rho \sigma} \partial_z \mathcal{H} \right). \quad (4)$$

Equation 4 is an expression of the CSC hypothesis, and this form will be assumed henceforth. The schematic in Figure 2 demonstrates the significance of this simple picture of cloud formation. Consider a hypothetical profile of clear-sky radiative cooling where the cooling rate decreases with height at two levels, $\partial_z \mathcal{H} < 0$ (Figure 2b), implying CSC at both levels (Figure 2a; Equation 4). Clouds must detrain at these heights to supply the mass demanded by the vertical divergence of clear air, implying that anvil clouds increase the areal cloud fraction. Tall, cumulonimbus towers with spreading anvils are thought to form by this mechanism, as depicted schematically in Figure 2a. We hypothesize that this same picture applies to congestus clouds and, if so, a decrease in clear-sky cooling must preferentially occur between 5-6 km by the action of one or more of Earth's greenhouse gases.

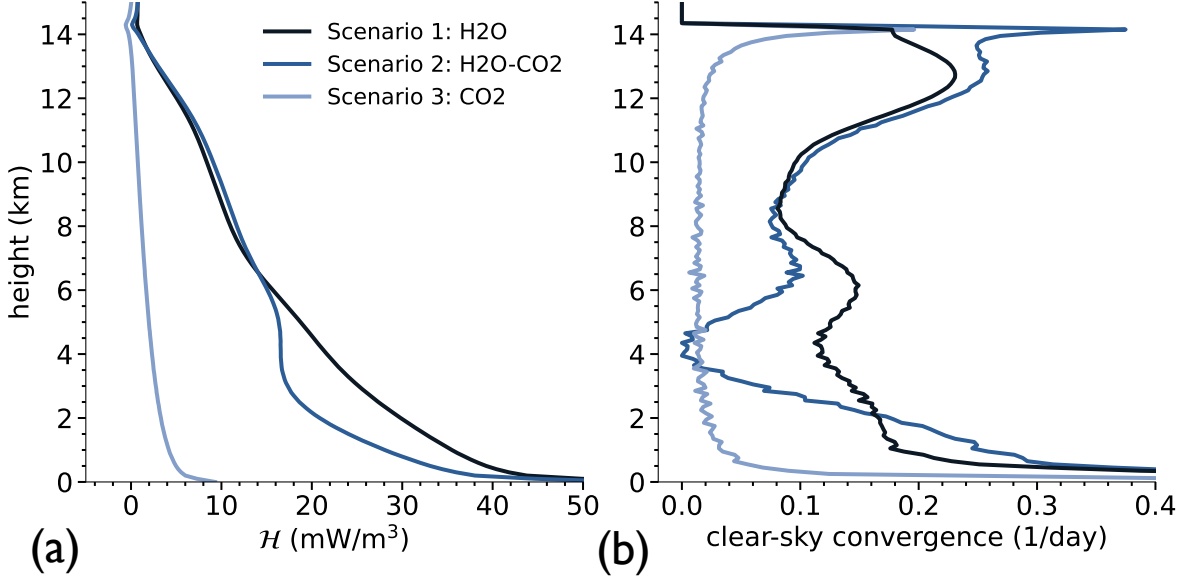


Figure 3. (a) Cooling rate $\mathcal{H}$ (Equation 5) from line-by-line computations with RFM and (b) clear-sky convergence (Equation 4) in three greenhouse gas scenarios.

3. CONGESTUS CLOUDS ARE ASSOCIATED WITH SPECTRAL FEATURES OF WATER VAPOR

Using the cooling profiles from the H2O, H2O-CO2, and CO2 scenarios (see Methods), we solve for the CSC using Equation 4 (Figure 3b). Below the tropopause, $\mathcal{H}$ monotonically decreases with height in all scenarios (Figure 3a). By comparing scenarios with and without water vapor, we see the well-known fact that water vapor accounts for the overwhelming majority of tropospheric cooling (Figure 3a; Held & Soden 2000). Figure 3b demonstrates that regions where $\mathcal{H}$ falls off fastest with height give rise to the peaks in CSC, in accordance with Equation 4. Importantly, the cooling rate in scenarios that include water vapor have a strong local decrease in the mid-troposphere that produces a peak in CSC at 6 km. The scenario without carbon dioxide does not have CSC in the mid-troposphere, indicating that the radiative properties of carbon dioxide do not contribute to CSC at these levels. H2O-CO2 demonstrates that the strong peak in mid-tropospheric CSC is not disrupted by the non-linear overlap of water vapor and carbon dioxide absorption lines. Our radiative calculations appear to validate the hypothesis for a CSC origin for observed tropical congestus cloud tops (Figure 1a).

To better understand the role of water vapor in driving CSC at 6 km, we resolve the spectral dimension of the cooling rate, $\mathcal{H}$ [Wm^{-3}]. $\mathcal{H}$ is, in fact, an integral over the spectrally-resolved cooling rate ($\tilde{\mathcal{H}}$; Wm^{-3}cm) in wavenumber space:

$$\mathcal{H}(z) = \int \tilde{\mathcal{H}}(\nu, z) d\nu \quad (5)$$

where ν is wavenumber [cm^{-1}] and z is height. We use tildes to identify any variable that is spectrally-resolved: i.e., a function of wavenumber. For now, we leave Equation 5 as an indefinite integral, and comment on the bounds of integration later. When we say colloquially that an atmospheric layer is radiatively cooling, we mean that more radiation is going out than coming in. In the case of a molecule like water vapor in Earth’s tropics, it is reasonable to make the cooling-to-space (CTS) approximation (Jeevanjee & Fueglistaler 2020b). In the CTS approximation, we assume that the non-local upwelling and downwelling fluxes cancel out, and so the net upward flux at any given height is equal to the fraction of the local blackbody emission that escapes to space (Jeevanjee & Fueglistaler 2020a).² We define the ratio of the emitted radiation-to-space to the blackbody radiation as the “emissivity-to-space” (Jeevanjee & Fueglistaler 2020a) and represent it symbolically as $\tilde{\epsilon}$. The CTS approximation isn’t necessary for our arguments, but it brings to the fore the underlying mechanism for CSC in Earth’s middle troposphere. The cooling rate in the

² In the SI, we verify that the CTS approximation agrees to within $\sim 10\%$ with $\mathcal{H}$ for water vapor and, more importantly, $\mathcal{H}_{cts}$ decreases rapidly with increasing height between 5-6 km.

CTS approximation may be expressed as a spectral integral over the product of two terms

$$\mathcal{H}_{cts}(z) = \int \pi \tilde{B} \times \partial_z \tilde{\epsilon} d\nu \quad (6)$$

(i) (ii)

where $\tilde{B}(\nu, T)$ is the Planck function for blackbody radiation. The terms in Equation 6 can be understood as follows: (i) $\pi \tilde{B}(\nu, T)$ is the blackbody emission at the local temperature T [$\text{Wm}^{-2}\text{cm}^{-1}$]³ and (ii) $\partial_z \tilde{\epsilon}$ is the emissivity-to-space gradient [m^{-1}]. $\tilde{\epsilon}$ is only a function of the optical depth,

$$\tilde{\tau} = \int_0^p \tilde{\kappa}(\nu, p, T) q dp/g, \quad (7)$$

which is, by convention, zero at the top of the atmosphere and increases monotonically towards the surface (Jeevanjee & Fueglistaler 2020b); and, in turn, depends almost entirely on pressure p , specific humidity q , and the absorption lines of water vapor $\tilde{\kappa}$. Since relative humidity is uniform and the lapse rate is constant in our “H2O” scenario, a swift decline (curvature) in the cooling rate can only emerge through $\tilde{\kappa}$.

Consider the absorption lines of water vapor at a reference temperature and pressure of 260 K and 500 hPa (Figure 4c), $\tilde{\kappa} = \tilde{\kappa}_{ref}(\nu, 500\text{hPa}, 260\text{K})$; absorption per unit mass is larger in the peaks than in the valleys. Figure 4a shows $\mathcal{H}_{cts}$ in H2O. It is clear that most of the cooling originates from the water vapor rotation band.⁴ Examining the tropospheric blackbody emission (Figure 4b) helps elucidate this phenomenon. The Earth’s tropospheric temperature range situates the emission peak roughly between 400-600 cm^{-1} , aligning the maximum values of $\pi \tilde{B}$ (those exceeding 0.3 $\text{Wm}^{-2}\text{cm}^{-1}$) with the water vapor rotation band. The non-uniform abundance of absorption lines in this band is the primary factor contributing to the strong curvature in the cooling rate. To substantiate this claim, we present 1D and 2D histograms depicting the spectral density of absorption lines (Figure 4d). Two distinct regions exhibit a significant drop in the line density. The first region, located near 40 m^2/kg (Jeevanjee & Fueglistaler 2020a), coincides with the upper-tropospheric peak in CSC associated with cumulonimbus anvils. The second region, spanning 0.3-1.1 m^2/kg , is associated with strong cooling between 5.2-6.9 km, as detailed in the Supplementary Information (SI). The spectral features of water vapor thus provide an important radiative constraint on the location of congestus cloud tops.

4. CONGESTUS INVIGORATION BY LOW RELATIVE HUMIDITY IN THE MID-TROPOSPHERE

A confounding property of spreading congestus cloud tops is that they occur in areas of low relative humidity (Figure 1), which is characteristic of the tropical mid troposphere. In fact, observations and cloud-resolving model simulations indicate that more mid-level clouds and fewer upper-level clouds occur in regions of relatively lower mid-tropospheric humidity (Redelsperger et al. 2002; Takemi et al. 2004; Kumar et al. 2014; Sokol & Hartmann 2022). This property could be partly responsible for the poor representation of the congestus cloud mode in global climate model simulations because cloud parameterizations are to some extent triggered by high environmental relative humidity. Near the surface in the planetary boundary layer, evaporation from the tropical oceans pins the relative humidity to values near saturation. The vertical distribution of relative humidity in the tropical atmosphere above the boundary layer is the result of a competition between subsidence drying and convective moistening (Romps 2014). The dominant mechanism switches from subsidence drying to convective moistening around 7.5 km in the tropics, producing the characteristic “C-shape” in tropical relative humidity (Figure 1e). In the previous section, we showed that water vapor spectral features constrain the height of tropical congestus clouds – recall that we arrived at this result using RFM in an idealized setup with a constant lapse rate and uniform relative humidity.

Under the conventional view, a decline in emissivity with height is required for anvil cloud formation (e.g., Hartmann & Larson 2002). We define an effective emissivity-to-space ϵ such that its vertical gradient is

$$\partial_z \epsilon(z) = \frac{\int_{150\text{cm}^{-1}}^{1000\text{cm}^{-1}} \pi \tilde{B} \times \partial_z \tilde{\epsilon} d\nu}{\int_{150\text{cm}^{-1}}^{1000\text{cm}^{-1}} \pi \tilde{B} d\nu}. \quad (8)$$

These terms should look familiar, as they also appear in Equation 6. The conventional view finds validation in Figure 5c where we see that the middle and upper layers of convective outflow coincide with the largest gradients in ϵ . In

³ The pre-factor of π comes from an integral of the spectral intensity over all upper-hemispheric solid angles (e.g., see Rybicki & Lightman 1985)

⁴ The rotation band is the spectral region to the left of $\sim 1000 \text{ cm}^{-1}$.

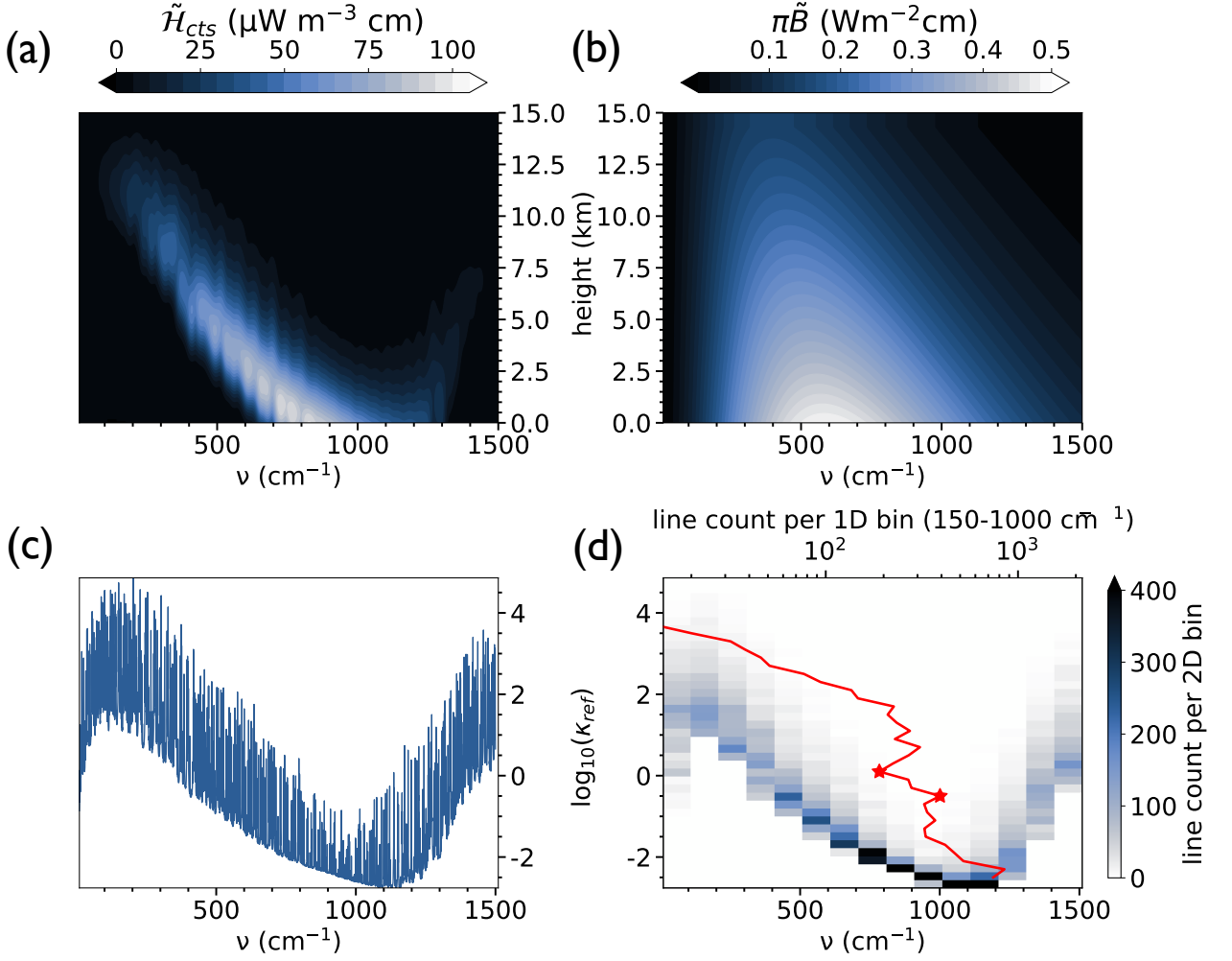


Figure 4. (a) Radiative cooling rate under the cooling-to-space approximation $\tilde{H}_{cts}$ (Equation 6) and (b) tropospheric blackbody emission $\pi\tilde{B}$. Both are resolved in wavenumber and height and vertically smoothed with a Gaussian filter. (c) Logarithm of the reference absorption coefficient of water vapor $\tilde{\kappa}_{ref}$ at 500 hPa and 260 K. (d) Two-dimensional (2D) histogram of absorption line abundance in 100 cm $^{-1}$ and 0.2 wavenumber- $\log_{10}(\tilde{\kappa}_{ref})$ bins. The red line is a one-dimensional (1D) histogram of absorption line abundance over a single spectral bin between 150-1000 cm $^{-1}$ in 0.2 increments of $\log_{10}(\tilde{\kappa}_{ref})$. Red stars mark a sharp drop in rotation-band line abundance between 0.3-1.1 m 2 /kg.

typical conditions where temperature decreases with height, a strong emissivity gradient tells us to expect strong CSC. *The mechanism that sets the height of congestus cloud tops is ultimately the same as that for the highest cloud anvils.*

Having argued for the spectral origin of congestus cloud tops, we next explore the effect of the “C-shape” in the vertical profile of relative humidity (Figure 5b) on the congestus mode. Figure 5d demonstrates that reducing the mid-level relative humidity locally enhances the CSC at middle levels and also reduces CSC at upper levels relative to the scenario with uniform relative humidity. Decreasing the relative humidity in this manner has two distinct radiative effects. First, it reduces the number of water vapor molecules at every level. This, in turn, forces the mid-level peak in CSC to descend from 6.0 km at 75% constant relative humidity to 5.3 km at 40% mid-tropospheric relative humidity (Figure 5d).⁵ The downward shift is explained by the fact that a drier troposphere increases the path length between the top of the atmosphere and the $\tilde{\tau} \approx 1$ levels in the water vapor rotation band, which determine where the strong gradients in radiative cooling, $\partial_z \mathcal{H}$, occur. Second, reducing the mid-tropospheric relative humidity accelerates the

⁵ 40% relative humidity in the mid-troposphere and a congestus peak of 5.3 km is more consistent with tropical reanalyses and observations (Figures 1d,e).

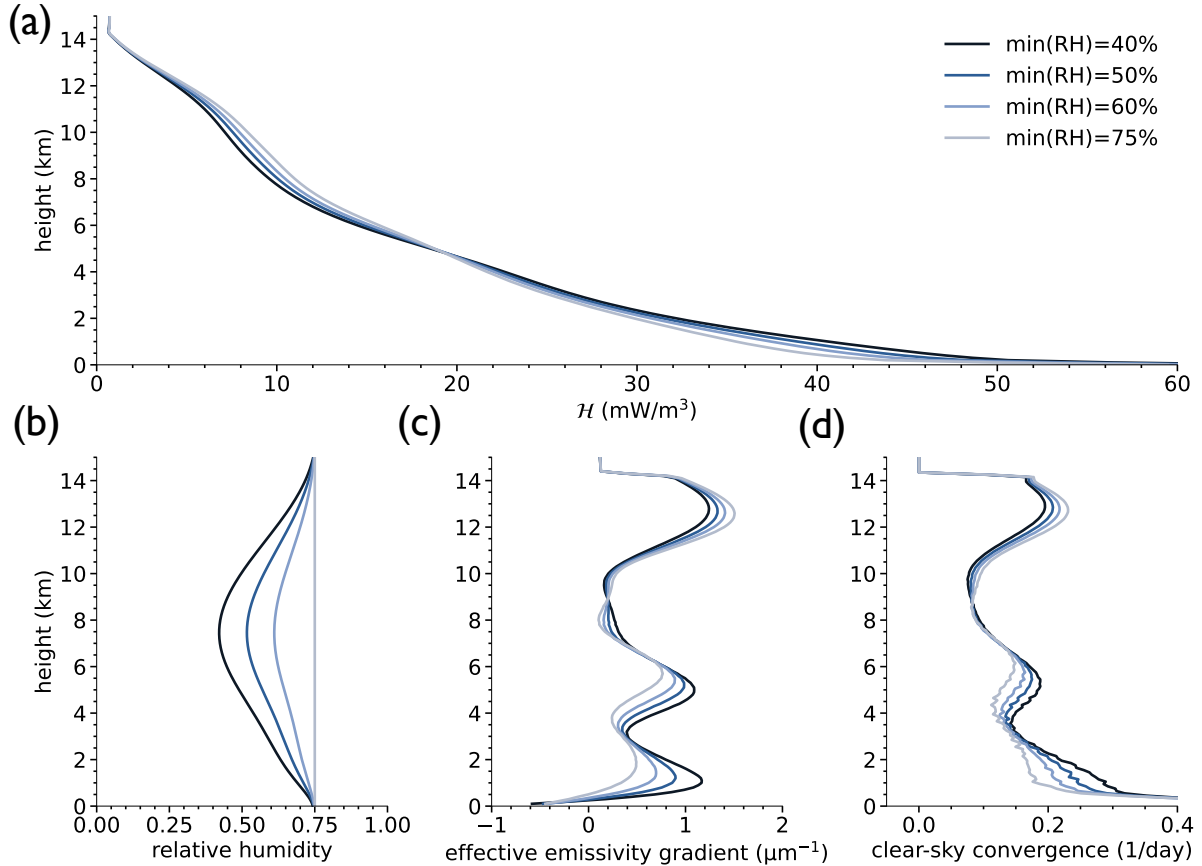


Figure 5. (a) Cooling rate $\mathcal{H}$ (Equation 5) from line-by-line computations with RFM in scenarios of varying (b) relative humidity. (c) Vertical gradient of effective emissivity (Equation 8) and (d) clear-sky convergence (Equation 4). The data are from H₂O and H₂O-C.

declining abundance of water molecules below 7.5 km, which increases the emissivity-to-space gradient (Figure 5c). This translates into a larger $\partial_z \mathcal{H}$ that in turn enhances the CSC in this region (Equation 4). In summary, a drier mid-troposphere promotes stronger CSC and this mechanism should lead to an enhancement of mid-level clouds.⁶

Our spectral interpretation of congestus enhancement is consistent with the invigoration of the congestus mode in cloud-resolving simulations in which convective aggregation emerges (Sokol & Hartmann 2022). The two consequences of aggregation that lead to congestus invigoration in their study are (1) a decrease in the mid- and upper-level relative humidity and (2) a stronger decrease in radiative cooling across the congestus level, which Sokol & Hartmann (2022) collectively refer to as the “radiative-divergence feedback”. Our line-by-line experiments with RFM bring additional clarity to the origin of this feedback. Congestus invigoration stems from two fundamental sources: (1) a drop in the spectral density of absorption lines in the water vapor rotation band that gives rise to a peak in CSC around 5-6 km and (2) a mid-level reduction in relative humidity that hastens the decline in $\mathcal{H}$ across the congestus level. Sokol & Hartmann (2022) find convective aggregation enhances the latter.

5. DISCUSSION

The processes that control the height and abundance of congestus clouds have been debated for some time. The debate has centered around the dry-air entrainment and the weak stability theories. In observational and modeling studies, these theories have been used successfully to show that the ascent of convective clouds can be stymied by buoyancy loss via entrainment and/or melting of cloud ice at the freezing level. These preceding theories individually

⁶ Decreasing the relative humidity in clear-sky regions is likely to reduce the cloud lifetime simultaneously (Seeley et al. 2019b), so this argument holds to the extent that increases in sources of cloud water dominate over the sinks.

fall short in offering a comprehensive explanation for two phenomena: first, that congestus clouds peak at 5-6 km and, second, that they are invigorated by low mid-tropospheric relative humidity. Therefore, the aim of this study has been to forge a complementary perspective of cloud formation that can explain a greater array of the observations.

The central result of this work is that a drop in the spectral density of absorption lines within the water vapor rotation band is the cause of a local, rapid decrease in the clear-sky cooling rate that, in turn, produces a peak in clear-sky convergence (CSC) between 5-6 km, a region associated with congestus cloud-top formation in the tropics (Wall et al. 2013). An equivalent statement is that congestus cloud tops form where there is a strong clear-sky gradient in the water vapor emissivity. The CSC hypothesis is derived from the basic building blocks of mass and energy conservation in radiative-convective equilibrium, which is approximately satisfied in the tropics over sufficiently large spatial and temporal scales (Jakob et al. 2019). It offers a simple, physical explanation for enhanced convective detrainment at specific heights of the atmosphere. We also demonstrated that a drier mid-troposphere creates larger gradients in water vapor emissivity that, in turn, enhance CSC, presumably driving more convective outflow at middle levels and less at upper levels, consistent with the observations. Our conclusions are based on theoretical considerations and results obtained from a line-by-line radiative transfer model.

However, there is still a need for a comprehensive assessment of our congestus formation theory using a cloud-resolving model. One possible path forward is through a mechanism-denial experiment. This could involve deliberately manipulating the $\tilde{\kappa}$ coefficient of water vapor to maintain a constant emission width across the spectral range linked to the congestus mode. If our spectral hypothesis for the origin of congestus clouds is correct, this alteration would lead to the disappearance of the peak in mid-level cloud fraction.

While it is well established that CSC supplies cloud water to clear-sky areas, it is important to note that our analysis does not also include sinks of cloud water such as precipitation and re-evaporation. The sink processes are thought to wield an equally influential role as the sources in shaping the time-mean cloud fraction (Seeley et al. 2019b; Beydoun et al. 2021; Jeevanjee & Zhou 2022). The lifetime of mid-level clouds, for example, is likely to shorten as the relative humidity decreases, in competition with the increase in CSC highlighted here (Figure 5d). Diagnosing the change in tropical cloud fraction as a function of environmental conditions such as surface temperature, carbon dioxide concentration, and mid-level relative humidity therefore requires an accurate accounting of both cloud sources and sinks. This task we leave to future work.

Finally, an important implication of the CSC hypothesis of congestus cloud top formation is that there is a *negative* correlation between mid-level relative humidity and mid-level cloudiness. However, in most, if not all, cloud schemes, a positive correlation between the relative humidity and cloudiness is assumed. This raises an important question: Can the representation of mid-level clouds be improved by designing parameterizations that take into account the observed anti-correlation between relative humidity and mid-level cloudiness? Our findings suggest this is a promising direction for future study.

6. METHODS

We employ the Reference Forward Model (RFM; Dudhia 2017) for line-by-line radiative transfer calculations at high resolution over a clear column of atmosphere. We use a vertical resolution of 100 m between the surface and the top of atmosphere at 30 km and a spectral resolution of 0.1 cm^{-1} between $10\text{--}1500 \text{ cm}^{-1}$. This spectral range was chosen because it includes 93% of the spectrally-integrated emission from a 300 K blackbody and 99% of the spectrally-integrated emission from a 200 K blackbody, the characteristic range of temperatures in the tropics. In our RFM calculations, we use the most recent line absorption data from the 2020 High-Resolution Transmission (HITRAN; Gordon et al. 2022) release. Continuum absorption by water vapor and carbon dioxide are included (where applicable) in our calculations, which are represented in RFM according to Mlawer et al. (2012) and Edwards (1992). Since oxygen and nitrogen are transparent to infrared radiation, we do not include them in our line-by-line calculations. As model inputs, we provide vertically-resolved profiles of temperature, pressure, and absorber mixing ratio that are representative of the tropical atmosphere. These inputs include a surface pressure and temperature of 1 bar and 300 K, a uniform lapse rate of 7 K/km ,⁷ and an isothermal stratosphere of 200 K. From these inputs, RFM computes the clear-sky infrared cooling rate, optical depth, transmissivity, and molecular absorption coefficients as a function of height, temperature, absorber concentration, and wavenumber. The line-by-line cooling rates from RFM are used to calculate the clear-sky convergence (Equation 4)

⁷ This is a simplifying assumption of constant stability σ (Equation 4). Note that σ decreases with height in typical conditions, which reduces the net CSC.

We construct four greenhouse gas scenarios for line-by-line radiative transfer calculations with RFM: (1) H₂O: Water vapor is the only absorber in this scenario. The troposphere has a uniform relative humidity of 75% and the stratospheric water vapor molar mixing ratio is pegged to the tropopause value; (2) H₂O-CO₂: This scenario includes carbon dioxide as well as water vapor as absorbing species. The molar mixing ratio of carbon dioxide is set to 350 ppmv at every height. Again, the relative humidity with respect to water vapor is 75% in the troposphere and, in the stratosphere, the water vapor molar mixing ratio is fixed to the tropopause value; (3) CO₂: Carbon dioxide is the sole absorber with a uniform mixing ratio of 350 ppmv; (4) H₂O-C: Water vapor is the only absorber. Relative humidity is allowed to vary with height. The shape of the profile is described in the main text. The stratosphere has a water vapor molar mixing ratio fixed at the tropopause value.

This work was supported by NSF Grant 1912673 and an Early-Career Fellowship from the Center for Diverse Leadership in Science at UCLA. The authors thank Anu Dudhia for technical assistance with RFM and William Bertrand and Jennifer Kay for sharing their CloudSat/CALIPSO dataset. The scientific colormaps used in our figures are from Crameri (2023).

The data and code from this study is freely available at 10.5281/zenodo.10139065.

REFERENCES

- Albrecht, B., Ghate, V., Mohrmann, J., et al. 2019, Bulletin of the American Meteorological Society, 100, 93, doi: [10.1175/BAMS-D-17-0180.1](https://doi.org/10.1175/BAMS-D-17-0180.1)
- Bertrand, W., Kay, J. E., Haynes, J., & de Boer, G. 2023, Earth System Science Data Discussions, 2023, 1, doi: [10.5194/essd-2023-265](https://doi.org/10.5194/essd-2023-265)
- Beydoun, H., Caldwell, P. M., Hannah, W. M., & Donahue, A. S. 2021, Geophysical Research Letters, 48, e2021GL094049, doi: <https://doi.org/10.1029/2021GL094049>
- Bodas-Salcedo, A., Webb, M. J., Brooks, M. E., et al. 2008, Journal of Geophysical Research: Atmospheres, 113, doi: <https://doi.org/10.1029/2007JD009620>
- Bourgeois, Q., Ekman, A. M. L., Igel, M. R., & Ringer, M. A. 2016, Nature Communications, 7, doi: <https://doi.org/10.1038/ncomms12432>
- Brown, R. G., & Zhang, C. 1997, Journal of the Atmospheric Sciences, 54, 2760, doi: [https://doi.org/10.1175/1520-0469\(1997\)054<2760:VOMMAI>2.0.CO;2](https://doi.org/10.1175/1520-0469(1997)054<2760:VOMMAI>2.0.CO;2)
- Cepi, P., & Nowack, P. 2021, Proceedings of the National Academy of Sciences, 118, e2026290118, doi: [10.1073/pnas.2026290118](https://doi.org/10.1073/pnas.2026290118)
- Cesana, G., Del Genio, A. D., & Chepfer, H. 2019, Earth System Science Data, 11, 1745, doi: [10.5194/essd-11-1745-2019](https://doi.org/10.5194/essd-11-1745-2019)
- Crameri, F. 2023, Scientific colour maps, Zenodo, doi: [10.5281/zenodo.1243862](https://doi.org/10.5281/zenodo.1243862)
- Dudhia, A. 2017, Journal of Quantitative Spectroscopy and Radiative Transfer, 186, 243, doi: <https://doi.org/10.1016/j.jqsrt.2016.06.018>
- Edwards, D. P. 1992, University Corporation for Atmospheric Research, (No. NCAR/TN-367+STR), doi: [10.5065/D6W37T86](https://doi.org/10.5065/D6W37T86)
- Gelaro, R., McCarty, W., Suárez, M. J., et al. 2017, Journal of Climate, 30, 5419, doi: <https://doi.org/10.1175/JCLI-D-16-0758.1>
- Global Modeling and Assimilation Office. 2015, MERRA-2 tavg3.3d.asn.Nv: 3d,3-Hourly,Time-Averaged,Model-Level,Assimilation,Assimilated Meteorological Fields V5.12.4, Greenbelt, MD, USA, Goddard Earth Sciences Data and Information Services Center (GES DISC), doi: [10.5067/SUOQESM06LPK](https://doi.org/10.5067/SUOQESM06LPK)
- Gordon, I., Rothman, L., Hargreaves, R., et al. 2022, Journal of Quantitative Spectroscopy and Radiative Transfer, 277, 107949, doi: <https://doi.org/10.1016/j.jqsrt.2021.107949>
- Hartmann, D. L., Dygert, B. D., Blossey, P. N., Fu, Q., & Sokol, A. B. 2022, Journal of Climate, 35, 6253, doi: <https://doi.org/10.1175/JCLI-D-21-0861.1>
- Hartmann, D. L., Gasparini, B., Berry, S. E., & Blossey, P. N. 2018, Journal of Advances in Modeling Earth Systems, 10, 3012, doi: <https://doi.org/10.1029/2018MS001484>
- Hartmann, D. L., & Larson, K. 2002, Geophysical Research Letters, 29, 12, doi: [10.1029/2002GL015835](https://doi.org/10.1029/2002GL015835)
- Held, I. M., & Soden, B. J. 2000, Annual Review of Energy and the Environment, 25, 441, doi: [10.1146/annurev.energy.25.1.441](https://doi.org/10.1146/annurev.energy.25.1.441)
- Hersbach, H., Bell, B., Berrisford, P., et al. 2020, Quarterly Journal of the Royal Meteorological Society, 146, 1999, doi: <https://doi.org/10.1002/qj.3803>

- , 2023, ERA5 monthly averaged data on single levels from 1940 to present, Copernicus Climate Change Service (C3S) Climate Data Store (CDS), doi: [10.24381/cds.fl7050d7](https://doi.org/10.24381/cds.fl7050d7)
- Hohenegger, C., & Stevens, B. 2013, *Journal of the Atmospheric Sciences*, 70, 448 , doi: <https://doi.org/10.1175/JAS-D-12-089.1>
- Jakob, C., Singh, M. S., & Jungandreas, L. 2019, *Journal of Geophysical Research: Atmospheres*, 124, 5418, doi: <https://doi.org/10.1029/2018JD030092>
- Jeevanjee, N., & Fueglistaler, S. 2020a, *Journal of the Atmospheric Sciences*, 77, 479 , doi: <https://doi.org/10.1175/JAS-D-18-0347.1>
- , 2020b, *Journal of the Atmospheric Sciences*, 77, 465 , doi: <https://doi.org/10.1175/JAS-D-18-0352.1>
- Jeevanjee, N., & Zhou, L. 2022, *Journal of Advances in Modeling Earth Systems*, 14, e2021MS002759, doi: <https://doi.org/10.1029/2021MS002759>
- Jensen, M. P., & Del Genio, A. D. 2006, *Journal of Climate*, 19, 2105 , doi: <https://doi.org/10.1175/JCLI3722.1>
- Johnson, R. H., Ciesielski, P. E., & Hart, K. A. 1996, *Journal of Atmospheric Sciences*, 53, 1838 , doi: [https://doi.org/10.1175/1520-0469\(1996\)053\(1838:TINTL\)2.0.CO;2](https://doi.org/10.1175/1520-0469(1996)053(1838:TINTL)2.0.CO;2)
- Johnson, R. H., Rickenbach, T. M., Rutledge, S. A., Ciesielski, P. E., & Schubert, W. H. 1999, *Journal of Climate*, 12, 2397 , doi: [https://doi.org/10.1175/1520-0442\(1999\)012\(2397:TCOTC\)2.0.CO;2](https://doi.org/10.1175/1520-0442(1999)012(2397:TCOTC)2.0.CO;2)
- Kuang, Z., & Hartmann, D. L. 2007, *Journal of Climate*, 20, 2051 , doi: <https://doi.org/10.1175/JCLI4124.1>
- Kumar, V. V., Protat, A., Jakob, C., & May, P. T. 2014, *Journal of the Atmospheric Sciences*, 71, 1105 , doi: <https://doi.org/10.1175/JAS-D-13-0231.1>
- Li, Y., Yang, P., North, G. R., & Dessler, A. 2012, *Journal of the Atmospheric Sciences*, 69, 2317 , doi: <https://doi.org/10.1175/JAS-D-11-0158.1>
- Mechem, D. B., & Oberthaler, A. J. 2013, *Journal of Advances in Modeling Earth Systems*, 5, 623, doi: <https://doi.org/10.1002/jame.20043>
- Miao, H., Wang, X., Liu, Y., & Wu, G. 2019, *Atmospheric Science Letters*, 20, e906, doi: <https://doi.org/10.1002/asl.906>
- Miyawaki, O., Shaw, T. A., & Jansen, M. F. 2022, *Journal of Climate*, 35, 1045 , doi: <https://doi.org/10.1175/JCLI-D-21-0440.1>
- Mlawer, E. J., Payne, V. H., Payne, V. H., et al. 2012, *Philos Trans A Math Phys Eng Sci*, 370, 2520, doi: [10.1098/rsta.2011.0295](https://doi.org/10.1098/rsta.2011.0295)
- Nuijens, L., & Emanuel, K. 2018, *Quarterly Journal of the Royal Meteorological Society*, 144, 2676, doi: <https://doi.org/10.1002/qj.3385>
- Petty, G. W. 1999, *Journal of Climate*, 12, 220 , doi: [https://doi.org/10.1175/1520-0442\(1999\)012\(0220:POPFWT\)2.0.CO;2](https://doi.org/10.1175/1520-0442(1999)012(0220:POPFWT)2.0.CO;2)
- Redelsperger, J.-L., Parsons, D. B., & Guichard, F. 2002, *Journal of the Atmospheric Sciences*, 59, 2438 , doi: [https://doi.org/10.1175/1520-0469\(2002\)059\(2438:RPAFLC\)2.0.CO;2](https://doi.org/10.1175/1520-0469(2002)059(2438:RPAFLC)2.0.CO;2)
- Riihimaki, L. D., McFarlane, S. A., & Comstock, J. M. 2012, *Journal of Climate*, 25, 6835 , doi: <https://doi.org/10.1175/JCLI-D-11-00599.1>
- Romps, D. M. 2014, *Journal of Climate*, 27, 7432, doi: [10.1175/JCLI-D-14-00255.1](https://doi.org/10.1175/JCLI-D-14-00255.1)
- Rybicki, G. B., & Lightman, A. P. 1985, *FUNDAMENTALS OF RADIATIVE TRANSFER* (John Wiley Sons, Ltd), 1–50, doi: <https://doi.org/10.1002/9783527618170.ch1>
- Sassen, K., Wang, Z., & Liu, D. 2009, *Journal of Geophysical Research: Atmospheres*, 114, doi: <https://doi.org/10.1029/2009JD011916>
- Seeley, J. T., Jeevanjee, N., Langhans, W., & Romps, D. M. 2019a, *Geophysical Research Letters*, 46, 492, doi: <https://doi.org/10.1029/2018GL080747>
- Seeley, J. T., Jeevanjee, N., & Romps, D. M. 2019b, *Geophysical Research Letters*, 46, 1842, doi: <https://doi.org/10.1029/2018GL080096>
- Seidel, S. D., & Yang, D. 2022, *Journal of Climate*, 35, 8065 , doi: <https://doi.org/10.1175/JCLI-D-21-0962.1>
- Singh, M. S., & O'Neill, M. E. 2022, *Rev. Mod. Phys.*, 94, 015001, doi: [10.1103/RevModPhys.94.015001](https://doi.org/10.1103/RevModPhys.94.015001)
- Smith, R. N. B. 1990, *Quarterly Journal of the Royal Meteorological Society*, 116, 435, doi: <https://doi.org/10.1002/qj.49711649210>
- Sokol, A. B., & Hartmann, D. L. 2022, *Journal of Advances in Modeling Earth Systems*, 14, e2022MS003045, doi: <https://doi.org/10.1029/2022MS003045>
- Spaulding-Astudillo, F. E., & Mitchell, J. L. 2023, *Journal of the Atmospheric Sciences*, doi: [10.1175/JAS-D-22-0063.1](https://doi.org/10.1175/JAS-D-22-0063.1)
- Sundqvist, H. 1988, *Parameterization of Condensation and Associated Clouds in Models for Weather Prediction and General Circulation Simulation*, ed. M. E. Schlesinger (Dordrecht: Springer Netherlands), 433–461, doi: [10.1007/978-94-009-3041-4_10](https://doi.org/10.1007/978-94-009-3041-4_10)
- Takayabu, Y. N., Shige, S., Tao, W.-K., & Hirota, N. 2010, *Journal of Climate*, 23, 2030 , doi: <https://doi.org/10.1175/2009JCLI3110.1>

- 489 Takemi, T., Hirayama, O., & Liu, C. 2004, Geophysical
 490 Research Letters, 31,
 491 doi: <https://doi.org/10.1029/2004GL020225>
- 492 Tompkins, A. 2005, The parametrization of cloud cover,
 493 ECMWF Moist Processes Lecture Note Series
- 494 Wall, C., Liu, C., & Zipser, E. 2013, Journal of Geophysical
 495 Research: Atmospheres, 118, 6478,
 496 doi: <https://doi.org/10.1002/jgrd.50455>
- 497 Zelinka, M. D., & Hartmann, D. L. 2010, Journal of
 498 Geophysical Research: Atmospheres, 115,
 499 doi: <https://doi.org/10.1029/2010JD013817>
- 500 Zuidema, P. 1998, Journal of the Atmospheric Sciences, 55,
 501 2220 ,
 502 doi: [https://doi.org/10.1175/1520-0469\(1998\)055<2220:](https://doi.org/10.1175/1520-0469(1998)055<2220:TMMITC>2.0.CO;2)
 503 [TMMITC>2.0.CO;2](https://doi.org/10.1175/1520-0469(1998)055<2220:TMMITC>2.0.CO;2)
- 504 Zuidema, P., Mapes, B., Lin, J., Fairall, C., & Wick, G.
 505 2006, Journal of Climate, 19, 4531 ,
 506 doi: <https://doi.org/10.1175/JCLI3836.1>

7. SUPPLEMENTARY INFORMATION

7.1. Spectral estimate of the height of congestus cloud tops

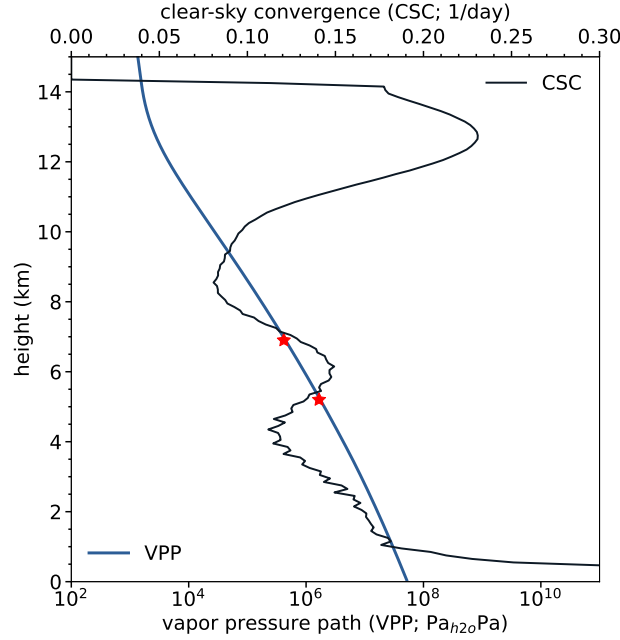


Figure 6. Clear-sky convergence in black. Vapor pressure path (Equation 11) in dark blue. The red stars denote the theoretical value of VPP at the mid-tropospheric peak in CSC (Equation 12). All data are from H₂O.

To connect with Hartmann et al. (2022), we derive the altitude of the mid-tropospheric peak in clear-sky convergence (CSC) using the vapor pressure path (VPP). $\mathcal{H}$ rapidly declines around 6 km due to a drop in the abundance of absorption lines in the water vapor rotation band between 0.3-1.1 m²/kg. These absorption lines are associated with strong cooling over a range of heights determined by their $\tilde{\tau} = 1$ levels. This information can be exploited to estimate the height of congestus cloud tops.

The absorption lines of water vapor are represented by the symbol $\tilde{\kappa}$. $\tilde{\kappa}$ is a fixed spectroscopic property of water vapor, and its value is uniquely determined as a function of wavenumber, pressure, and temperature.

$$\tilde{\kappa}(\nu, p, T) \approx \tilde{\kappa}(\nu, p) = \tilde{\kappa}_{ref}(\nu, 500\text{hPa}, 260\text{K}) \frac{p}{p_{ref}} D. \quad (9)$$

$\tilde{\kappa}_{ref}$ is the absorption coefficient distribution at a reference temperature (T_{ref}) and pressure (p_{ref}) of 260 K and 500 hPa. The roughly linear dependence of $\tilde{\kappa}$ on pressure is given by the factor p/p_{ref} , and results from pressure broadening of absorption lines. Note that, for analytic tractability, we neglect the minor dependence of $\tilde{\kappa}$ on temperature. We use the default RFM value for the two-stream diffusivity factor $D = 1.5$.

To retrieve the congestus peak, we simply solve for the height of maximum cooling associated with the absorption lines $\tilde{\kappa}_{ref} = 0.3\text{-}1.1$ m²/kg. Using an approximation for the specific humidity,

$$q \approx \frac{R_d}{R_v} \frac{e}{p}, \quad (10)$$

defining the vapor pressure path (Hartmann et al. 2022) as

$$\text{VPP} = \int_0^p e \, dp, \quad (11)$$

and setting $\tilde{\tau} = 1$ (Equation 7), we arrive at a theoretical prediction for the VPP at the congestus level:

$$\text{VPP} = \frac{g R_v p_{ref}}{D R_d \tilde{\kappa}_{ref}}, \quad (12)$$

where e is the partial pressure of water vapor and R_d and R_v are the specific gas constants of dry air and water vapor, respectively. Incredibly, the VPP is a *conserved quantity* of congestus clouds because it is only a function of thermodynamic and spectroscopic constants. We solve Equation 12 with the idealized profiles of temperature, pressure, and humidity constructed for RFM (see Methods). This yields a height of 5.2-6.9 km for the mid-tropospheric peak in CSC, as expected from Figure 6.

7.2. Validation of the cooling-to-space approximation for water vapor

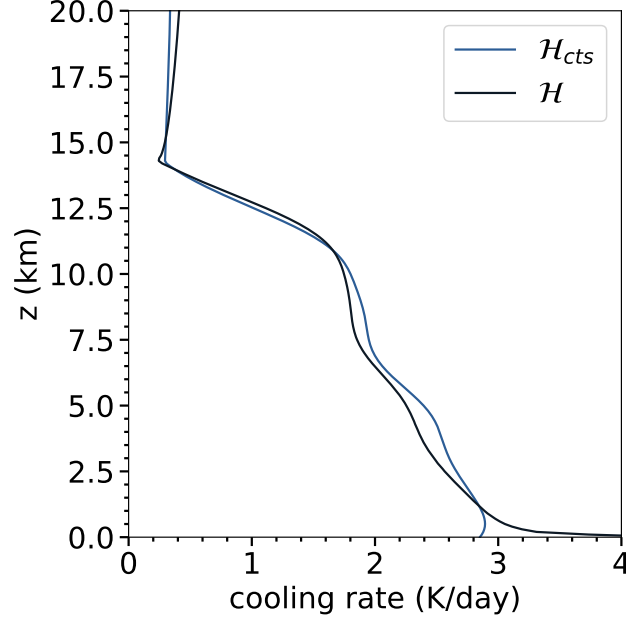


Figure 7. Comparison of the total cooling rate (black) to the cooling-to-space rate (blue) in H₂O.

Figure 7 compares the full cooling rate $\mathcal{H}$ to the cooling-to-space rate $\mathcal{H}_{cts}$ from H₂O. Note that the cooling rate is given in units of K/day. The main takeaway is that $\mathcal{H}_{cts}$ is in excellent agreement with $\mathcal{H}$. Both $\mathcal{H}$ and $\mathcal{H}_{cts}$ exhibit a local rapid decrease around 6 km, justifying our choice to base our analysis on $\mathcal{H}_{cts}$.